

Non-local mathematical physics

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1 Differential equations as linear operators

1.1 Linear operators and the superposition principle

A (differential) operator L is **linear** if

$$L(a\phi + b\psi) = aL(\phi) + bL(\psi), \quad a, b \in \mathbb{R} \text{ or } \mathbb{C}. \quad (1)$$

A linear ODE/PDE can be written abstractly as

$$L\psi = F, \quad (2)$$

where F is a known “source” function and ψ is the unknown solution. For the homogeneous equation

$$L\psi = 0, \quad (3)$$

the **superposition principle** holds: if ψ_1, ψ_2 solve $L\psi = 0$, then $c_1\psi_1 + c_2\psi_2$ also solves it.

1.2 Second-order linear ODE as an operator equation

A general second-order linear differential operator can be written as

$$Lu(x) = p_0(x)u''(x) + p_1(x)u'(x) + p_2(x)u(x), \quad (4)$$

with real coefficient functions p_0, p_1, p_2 (smooth enough on the interval of interest).

1.3 Eigenvalue problems

Many physics problems are eigenvalue problems:

$$Lu + \lambda w(x)u = 0, \quad w(x) > 0 \text{ (except possibly isolated points)}. \quad (5)$$

A nontrivial solution u_λ that satisfies the required boundary conditions is called an **eigenfunction** and the corresponding λ is an **eigenvalue**. Typically, boundary conditions restrict λ to a **discrete spectrum**.

Quantum mechanics example. The time-independent Schrödinger equation is an eigenvalue equation:

$$H\psi = E\psi, \quad (6)$$

where H is the Hamiltonian operator and E is the energy eigenvalue.

2 Self-adjoint ODEs (Sturm–Liouville form)

2.1 Adjoint operator and self-adjointness

Define an inner-product-like bilinear form (real case for simplicity)

$$\langle u, Lu \rangle := \int_a^b u(x) Lu(x) dx. \quad (7)$$

Starting from

$$Lu = p_0 u'' + p_1 u' + p_2 u, \quad (8)$$

one defines an **adjoint operator** $\bar{L}$ such that (schematically)

$$\int_a^b v Lu \, dx = \int_a^b (\bar{L}v) u \, dx \quad (\text{plus boundary terms}). \quad (9)$$

If the boundary terms vanish under the imposed boundary conditions and

$$\bar{L} = L, \quad (10)$$

then L is called **self-adjoint**.

2.2 Self-adjoint (Sturm–Liouville) operator form

A second-order differential operator is in **self-adjoint form** if it can be written as

$$Lu = \frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + q(x) u(x), \quad (11)$$

with real functions $p(x), q(x)$.

2.3 Converting a general ODE to self-adjoint form

Given

$$Lu = p_0 u'' + p_1 u' + p_2 u, \quad (12)$$

one can multiply by a suitable integrating factor to obtain a self-adjoint form. Define

$$f(x) = \exp \left(\int^x \frac{p_1(t)}{p_0(t)} dt \right), \quad (13)$$

then

$$\frac{f(x)}{p_0(x)} Lu = \frac{d}{dx} \left(f(x) \frac{du}{dx} \right) + \frac{p_2(x)}{p_0(x)} f(x) u, \quad (14)$$

which is self-adjoint.

2.4 Sturm–Liouville eigenvalue problem

The standard Sturm–Liouville problem is

$$\frac{d}{dx} \left(p(x) \frac{du}{dx} \right) + q(x) u + \lambda w(x) u = 0, \quad a \leq x \leq b, \quad (15)$$

with boundary conditions chosen so that the operator is Hermitian/self-adjoint.

Important note. Self-adjointness is not only about the differential expression: it depends crucially on the **function space** and **boundary conditions**.

3 Inner product, orthogonality, and normalization

3.1 Weighted inner product

For Sturm–Liouville problems the natural inner product is

$$\langle v|u \rangle := \int_a^b v^*(x) w(x) u(x) dx, \quad (16)$$

where $w(x)$ is the weight function.

3.2 Orthogonality of eigenfunctions

Let u_i, u_j satisfy

$$Lu_i + \lambda_i w u_i = 0, \quad Lu_j + \lambda_j w u_j = 0. \quad (17)$$

If L is Hermitian (with boundary conditions) then for $\lambda_i \neq \lambda_j$:

$$\int_a^b u_i(x) u_j^*(x) w(x) dx = 0. \quad (18)$$

So eigenfunctions with distinct eigenvalues are orthogonal under the weight w .

3.3 Normalization

If u_n is an eigenfunction, scaling is arbitrary for homogeneous equations:

$$u_n \rightarrow c u_n. \quad (19)$$

We often choose orthonormality:

$$\int_a^b u_m(x) u_n^*(x) w(x) dx = \delta_{mn}. \quad (20)$$

4 Hermitian operators: reality of eigenvalues

4.1 Hermitian condition

An operator L is **Hermitian** if

$$\int \psi_1^*(x) L\psi_2(x) dx = \int (L\psi_1(x))^* \psi_2(x) dx \quad (21)$$

for all functions in the domain satisfying the boundary conditions.

4.2 Eigenvalues are real

If

$$Lu_i + \lambda_i w u_i = 0 \quad (22)$$

and L is Hermitian, then $\lambda_i \in \mathbb{R}$. Sketch: multiply the equation by u_i^* , subtract the conjugate equation times u_i , integrate, and use the Hermitian property to force

$$\lambda_i^* = \lambda_i. \quad (23)$$

4.3 Expectation values in quantum mechanics

For a Hermitian operator L and wavefunction ψ ,

$$\langle L \rangle := \int \psi^*(x) L\psi(x) dx \quad (24)$$

is real, even if ψ is not an eigenfunction.

5 Uniqueness, linear independence, and degeneracy

5.1 Uniqueness for second-order linear ODEs (idea)

For a second-order linear homogeneous ODE, the solution space is 2-dimensional: there are exactly two linearly independent solutions locally (given appropriate regularity). Imposing two boundary/initial conditions typically selects a unique solution.

5.2 Linear independence of eigenfunctions (distinct eigenvalues)

If L has eigenfunctions u_1, u_2 with distinct eigenvalues $\lambda_1 \neq \lambda_2$, then u_1 and u_2 must be linearly independent. Reason: if $u_2 = c u_1$, then applying L forces $\lambda_2 = \lambda_1$.

5.3 Degeneracy

An eigenvalue λ is **N -fold degenerate** if there exist N linearly independent eigenfunctions corresponding to the same eigenvalue.

Key point. If $\lambda_i = \lambda_j$ (degenerate case), orthogonality is *not automatic*. The Sturm–Liouville orthogonality proof only forces orthogonality for $\lambda_i \neq \lambda_j$.

5.4 Orthogonalizing degenerate eigenfunctions

Even if eigenfunctions are degenerate, one can always choose an orthogonal basis within the degenerate subspace using Gram–Schmidt.

5.5 Degeneracy from symmetry (physics intuition)

Degeneracy is often caused by symmetry. Example: hydrogen atom energy depends on n but not on (ℓ, m) in the Coulomb potential, leading to an n^2 -fold degeneracy (ignoring spin). Breaking symmetry (e.g. magnetic field) can split levels (Zeeman effect).

6 Concrete example: Fourier eigenfunctions (periodic interval)

Consider

$$\frac{d^2 u}{dx^2} + n^2 u = 0, \quad (25)$$

on an interval of length 2π with periodic boundary conditions. Eigenfunctions:

$$u_n(x) = \cos(nx), \quad v_n(x) = \sin(nx), \quad (26)$$

and the eigenvalue is $\lambda = n^2$. This eigenvalue is **two-fold degenerate** for $n \geq 1$ because both $\sin(nx)$ and $\cos(nx)$ correspond to the same n^2 .

Orthogonality relations (on $[x_0, x_0 + 2\pi]$):

$$\int_{x_0}^{x_0+2\pi} \sin(mx) \sin(nx) dx = C_n \delta_{mn}, \quad (27)$$

$$\int_{x_0}^{x_0+2\pi} \cos(mx) \cos(nx) dx = D_n \delta_{mn}, \quad (28)$$

$$\int_{x_0}^{x_0+2\pi} \sin(mx) \cos(nx) dx = 0. \quad (29)$$

7 What to remember (high-yield summary)

- Linear DEs can be written as $L\psi = F$; if $F = 0$ then superposition holds.
- Self-adjoint form is

$$Lu = (pu')' + qu.$$

- Sturm–Liouville eigenvalue equation:

$$(pu')' + qu + \lambda wu = 0.$$

- Weighted inner product:

$$\langle v|u \rangle = \int_a^b v^* w u dx.$$

- If L is Hermitian/self-adjoint (with boundary conditions):
 - eigenvalues are real,
 - eigenfunctions with distinct eigenvalues are orthogonal,
 - degeneracy means same eigenvalue can have multiple independent eigenfunctions.
- Degenerate eigenfunctions are not automatically orthogonal, but can be orthogonalized.

8 Graham Schmidt

9 Gram–Schmidt orthogonalization of functions

9.1 Goal and setting

Given a set of **linearly independent** functions

$$\{f_1(x), f_2(x), \dots, f_N(x)\}$$

that are not necessarily orthogonal, we want to construct an **orthogonal** (or orthonormal) set

$$\{\phi_1(x), \phi_2(x), \dots, \phi_N(x)\}$$

that spans the same subspace.

9.2 Inner product and norm (weighted)

In Sturm–Liouville theory the natural inner product is

$$\langle g|f \rangle := \int_a^b g^*(x) w(x) f(x) dx, \quad w(x) > 0. \quad (30)$$

The corresponding norm is

$$\|f\| := \sqrt{\langle f|f \rangle}. \quad (31)$$

Orthogonality means

$$\langle \phi_m|\phi_n \rangle = 0, \quad m \neq n. \quad (32)$$

Orthonormality means

$$\langle \phi_m|\phi_n \rangle = \delta_{mn}. \quad (33)$$

9.3 Gram–Schmidt construction (orthogonal set)

Define recursively:

$$u_1 := f_1, \quad (34)$$

$$u_2 := f_2 - \frac{\langle u_1|f_2 \rangle}{\langle u_1|u_1 \rangle} u_1, \quad (35)$$

$$u_3 := f_3 - \frac{\langle u_1|f_3 \rangle}{\langle u_1|u_1 \rangle} u_1 - \frac{\langle u_2|f_3 \rangle}{\langle u_2|u_2 \rangle} u_2, \quad (36)$$

$$\vdots \quad (37)$$

$$u_n := f_n - \sum_{k=1}^{n-1} \frac{\langle u_k|f_n \rangle}{\langle u_k|u_k \rangle} u_k. \quad (38)$$

Then the set $\{u_1, \dots, u_N\}$ is orthogonal:

$$\langle u_m|u_n \rangle = 0 \quad (m \neq n).$$

9.4 Compact projection form

Define the projection of f onto u_k as

$$\text{proj}_{u_k}(f) := \frac{\langle u_k | f \rangle}{\langle u_k | u_k \rangle} u_k. \quad (39)$$

Then Gram–Schmidt is simply

$$u_n = f_n - \sum_{k=1}^{n-1} \text{proj}_{u_k}(f_n). \quad (40)$$

9.5 Orthonormalization step

To normalize each u_n , define

$$\phi_n := \frac{u_n}{\|u_n\|} = \frac{u_n}{\sqrt{\langle u_n | u_n \rangle}}. \quad (41)$$

Then

$$\langle \phi_m | \phi_n \rangle = \delta_{mn}.$$

9.6 Why Gram–Schmidt works (key identity)

At step n , we subtract off all components of f_n parallel to the already-built orthogonal vectors $u_1, \dots, u_{n-1}$. Thus u_n lies in the span of $\{f_1, \dots, f_n\}$ but is orthogonal to the previous ones.

Specifically, for any $j < n$:

$$\langle u_j | u_n \rangle = \left\langle u_j \left| f_n - \sum_{k=1}^{n-1} \frac{\langle u_k | f_n \rangle}{\langle u_k | u_k \rangle} u_k \right. \right\rangle \quad (42)$$

$$= \langle u_j | f_n \rangle - \sum_{k=1}^{n-1} \frac{\langle u_k | f_n \rangle}{\langle u_k | u_k \rangle} \langle u_j | u_k \rangle \quad (43)$$

$$= \langle u_j | f_n \rangle - \frac{\langle u_j | f_n \rangle}{\langle u_j | u_j \rangle} \langle u_j | u_j \rangle \quad (44)$$

$$= 0. \quad (45)$$

9.7 Important practical notes

- **Linear independence required:** If $\{f_n\}$ are linearly dependent, then some u_n becomes the zero function ($\|u_n\| = 0$) and the algorithm breaks.
- **Order matters:** Different ordering of $\{f_n\}$ produces different orthogonal bases.
- **Same subspace:** For each n ,

$$\text{span}\{u_1, \dots, u_n\} = \text{span}\{f_1, \dots, f_n\}.$$

- **Degenerate eigenspaces:** If an eigenvalue is degenerate, Gram–Schmidt produces an orthonormal basis inside the degenerate subspace.

9.8 Two-function special case (fast formula)

Given two linearly independent functions f_1, f_2 :

$$u_1 = f_1, \quad (46)$$

$$u_2 = f_2 - \frac{\langle f_1 | f_2 \rangle}{\langle f_1 | f_1 \rangle} f_1, \quad (47)$$

$$\phi_1 = \frac{f_1}{\|f_1\|}, \quad \phi_2 = \frac{u_2}{\|u_2\|}. \quad (48)$$

9.9 Example (typical weight)

On $[-1, 1]$ with weight $w(x) = 1$:

$$\langle g | f \rangle = \int_{-1}^1 g(x) f(x) dx.$$

Start with non-orthogonal polynomials:

$$f_1(x) = 1, \quad f_2(x) = x, \quad f_3(x) = x^2.$$

Then

$$u_1 = 1, \quad u_2 = x - \frac{\langle 1 | x \rangle}{\langle 1 | 1 \rangle} \cdot 1 = x,$$

since $\int_{-1}^1 x dx = 0$. Next,

$$u_3 = x^2 - \frac{\langle 1 | x^2 \rangle}{\langle 1 | 1 \rangle} \cdot 1 - \frac{\langle x | x^2 \rangle}{\langle x | x \rangle} \cdot x.$$

But $\int_{-1}^1 x^3 dx = 0$, so the x -projection vanishes and

$$u_3 = x^2 - \frac{\int_{-1}^1 x^2 dx}{\int_{-1}^1 1 dx} = x^2 - \frac{2/3}{2} = x^2 - \frac{1}{3}.$$

These are proportional to the first Legendre polynomials:

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1).$$

9.10 Final high-yield summary

- Gram–Schmidt turns $\{f_n\}$ into orthogonal $\{u_n\}$ by subtracting projections.
- Weighted inner product:

$$\langle g | f \rangle = \int_a^b g^* w f dx.$$

- Core recursion:

$$u_n = f_n - \sum_{k=1}^{n-1} \frac{\langle u_k | f_n \rangle}{\langle u_k | u_k \rangle} u_k.$$

- Normalize:

$$\phi_n = \frac{u_n}{\sqrt{\langle u_n | u_n \rangle}}.$$

- Used to build orthonormal bases in degenerate eigenspaces.

10 Integral Fourier transforms

10.1 Fourier transform pair (physics convention)

Let $f(x)$ be integrable (or in a suitable distribution sense). Define the Fourier transform:

$$\tilde{f}(k) = \mathcal{F}\{f\}(k) := \int_{-\infty}^{\infty} f(x) e^{-ikx} dx. \quad (49)$$

The inverse transform is

$$f(x) = \mathcal{F}^{-1}\{\tilde{f}\}(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk. \quad (50)$$

Interpretation. The Fourier transform expands $f(x)$ in the continuous basis of plane waves e^{ikx} .

10.2 Basic transform properties

Linearity.

$$\mathcal{F}\{af + bg\} = a\tilde{f} + b\tilde{g}. \quad (51)$$

Shift in x (translation). If $g(x) = f(x - x_0)$ then

$$\tilde{g}(k) = e^{-ikx_0} \tilde{f}(k). \quad (52)$$

Modulation (shift in k). If $g(x) = e^{ik_0x} f(x)$ then

$$\tilde{g}(k) = \tilde{f}(k - k_0). \quad (53)$$

Scaling. If $g(x) = f(ax)$ with $a \neq 0$ then

$$\tilde{g}(k) = \frac{1}{|a|} \tilde{f}\left(\frac{k}{a}\right). \quad (54)$$

11 Dirac delta function and its Fourier representation

11.1 Defining property

The Dirac delta $\delta(x)$ is defined by

$$\int_{-\infty}^{\infty} \delta(x - a) f(x) dx = f(a), \quad (55)$$

for suitable test functions f .

11.2 Fourier representation of δ

Start from the inverse transform of $\tilde{f}(k) = 1$:

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk. \quad (56)$$

More generally,

$$\delta(x - x_0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-x_0)} dk. \quad (57)$$

11.3 Delta as completeness / orthogonality of plane waves

Using the transform pair,

$$\int_{-\infty}^{\infty} e^{ikx} dx = 2\pi \delta(k), \quad (58)$$

and equivalently,

$$\int_{-\infty}^{\infty} e^{i(k-k')x} dx = 2\pi \delta(k - k'). \quad (59)$$

11.4 Fourier transform of a delta

$$\mathcal{F}\{\delta(x - x_0)\}(k) = \int_{-\infty}^{\infty} \delta(x - x_0) e^{-ikx} dx = e^{-ikx_0}. \quad (60)$$

12 Fourier transforms of derivatives

Assume boundary terms vanish (e.g. $f(x) \rightarrow 0$ sufficiently fast as $|x| \rightarrow \infty$), or interpret in the distribution sense.

12.1 First derivative

$$\mathcal{F}\{f'(x)\}(k) = \int_{-\infty}^{\infty} f'(x) e^{-ikx} dx \quad (61)$$

$$= \left[f(x) e^{-ikx} \right]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} f(x) e^{-ikx} dx \quad (62)$$

$$= ik \tilde{f}(k). \quad (63)$$

So:

$$\boxed{\mathcal{F}\{f'\}(k) = ik \tilde{f}(k)}. \quad (64)$$

12.2 Second derivative

$$\mathcal{F}\{f''\}(k) = (ik)^2 \tilde{f}(k) = -k^2 \tilde{f}(k). \quad (65)$$

12.3 n th derivative

$$\boxed{\mathcal{F}\left\{\frac{d^n f}{dx^n}\right\}(k) = (ik)^n \tilde{f}(k)}. \quad (66)$$

Key use in PDEs. Differential operators become algebraic multipliers in Fourier space:

$$\frac{d}{dx} \rightarrow ik, \quad \frac{d^2}{dx^2} \rightarrow -k^2.$$

13 Convolution theorem

13.1 Definition of convolution

For functions f, g define their convolution

$$(f * g)(x) := \int_{-\infty}^{\infty} f(x-t) g(t) dt = \int_{-\infty}^{\infty} f(t) g(x-t) dt. \quad (67)$$

13.2 Fourier transform of a convolution

$$\boxed{\mathcal{F}\{(f * g)(x)\}(k) = \tilde{f}(k) \tilde{g}(k)}. \quad (68)$$

13.3 Proof sketch (one-line computation)

$$\mathcal{F}\{f * g\}(k) = \int dx e^{-ikx} \int dt f(x-t) g(t) \quad (69)$$

$$= \int dt g(t) \int du f(u) e^{-ik(u+t)} \quad (70)$$

$$= \left(\int du f(u) e^{-iku} \right) \left(\int dt g(t) e^{-ikt} \right) \quad (71)$$

$$= \tilde{f}(k) \tilde{g}(k). \quad (72)$$

13.4 Inverse form: product becomes convolution

If $h(x) = f(x)g(x)$ then

$$\boxed{\tilde{h}(k) = \mathcal{F}\{fg\}(k) = \frac{1}{2\pi} (\tilde{f} * \tilde{g})(k)}. \quad (73)$$

13.5 Delta as identity under convolution

Using δ ,

$$(f * \delta)(x) = \int f(x-t) \delta(t) dt = f(x). \quad (74)$$

14 High-yield summary

- Transform pair:

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk.$$

- Delta representation:

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk, \quad \int e^{i(k-k')x} dx = 2\pi \delta(k - k').$$

- Derivatives:

$$\mathcal{F}\{f^{(n)}\}(k) = (ik)^n \tilde{f}(k).$$

- Convolution theorem:

$$\mathcal{F}\{f * g\} = \tilde{f} \tilde{g}, \quad \mathcal{F}\{fg\} = \frac{1}{2\pi}(\tilde{f} * \tilde{g}).$$

15 Laplace transforms

15.1 Definition

For a function $f(t)$ (typically $t \geq 0$), the Laplace transform is

$$F(s) = \mathcal{L}\{f(t)\}(s) := \int_0^\infty e^{-st} f(t) dt, \quad s \in \mathbb{C}. \quad (75)$$

The integral converges in a half-plane

$$\operatorname{Re}(s) > s_0$$

determined by the growth of $f(t)$ (“region of convergence”).

Linearity.

$$\mathcal{L}\{af + bg\} = a\mathcal{L}\{f\} + b\mathcal{L}\{g\}. \quad (76)$$

15.2 Useful basic transforms

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad (77)$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 0, 1, 2, \dots \quad (78)$$

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad (79)$$

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2 + \omega^2}, \quad (80)$$

$$\mathcal{L}\{\sin(\omega t)\} = \frac{\omega}{s^2 + \omega^2}. \quad (81)$$

16 Laplace transform of derivatives

Assume $f(t)$ is sufficiently smooth and of exponential order so that the transforms exist.

16.1 First derivative

$$\mathcal{L}\{f'(t)\}(s) = \int_0^\infty e^{-st} f'(t) dt \quad (82)$$

$$= \left[e^{-st} f(t) \right]_0^\infty + s \int_0^\infty e^{-st} f(t) dt \quad (83)$$

$$= -f(0) + sF(s). \quad (84)$$

Hence the key identity:

$$\boxed{\mathcal{L}\{f'(t)\}(s) = sF(s) - f(0).} \quad (85)$$

16.2 Second derivative

Apply the first-derivative rule to $f'(t)$:

$$\boxed{\mathcal{L}\{f''(t)\}(s) = s^2 F(s) - sf(0) - f'(0).} \quad (86)$$

16.3 n th derivative

By induction:

$$\boxed{\mathcal{L}\{f^{(n)}(t)\}(s) = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0).} \quad (87)$$

High-yield use. Laplace transforms turn ODE initial-value problems into algebra:

$$\text{ODE in } t \longrightarrow \text{algebraic equation in } s.$$

17 Laplace transform of Dirac delta and Heaviside functions

17.1 Dirac delta $\delta(t - a)$

Using the defining property of the delta distribution,

$$\int_0^\infty \delta(t - a) g(t) dt = g(a), \quad a > 0,$$

we obtain

$$\boxed{\mathcal{L}\{\delta(t - a)\}(s) = \int_0^\infty e^{-st} \delta(t - a) dt = e^{-as}.} \quad (88)$$

Special case:

$$\mathcal{L}\{\delta(t)\}(s) = 1. \quad (89)$$

17.2 Heaviside step function $H(t - a)$

Define

$$H(t - a) = \begin{cases} 0, & t < a, \\ 1, & t > a. \end{cases}$$

Then

$$\mathcal{L}\{H(t - a)\}(s) = \int_0^\infty e^{-st} H(t - a) dt = \int_a^\infty e^{-st} dt \quad (90)$$

$$= \boxed{\frac{e^{-as}}{s}}. \quad (91)$$

17.3 Time-shift theorem (with Heaviside)

If $g(t) = H(t - a) f(t - a)$, then

$$\boxed{\mathcal{L}\{H(t - a)f(t - a)\}(s) = e^{-as}F(s).} \quad (92)$$

This is the standard tool for piecewise forcing functions.

18 Convolution theorem (Laplace)

18.1 Definition of convolution on $[0, \infty)$

For functions f, g on $t \geq 0$, define

$$(f * g)(t) := \int_0^t f(\tau) g(t - \tau) d\tau. \quad (93)$$

18.2 Laplace transform of convolution

$$\boxed{\mathcal{L}\{(f * g)(t)\}(s) = F(s) G(s).} \quad (94)$$

18.3 Inverse statement

$$\boxed{\mathcal{L}^{-1}\{F(s)G(s)\}(t) = (f * g)(t).} \quad (95)$$

High-yield use. Convolution appears naturally when solving linear ODEs/PDEs with forcing and in Green's function methods.

19 Inverse Laplace transform and Bromwich integral

19.1 Inverse Laplace transform (Bromwich inversion)

The inverse Laplace transform is given by the Bromwich integral:

$$\boxed{f(t) = \mathcal{L}^{-1}\{F(s)\}(t) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} e^{st} F(s) ds.} \quad (96)$$

Here γ is chosen so that the vertical line $\text{Re}(s) = \gamma$ lies in the region of convergence and to the right of all singularities of $F(s)$.

19.2 Contour interpretation (complex analysis link)

The Bromwich integral is evaluated using residue calculus by closing the contour (usually to the left half-plane for $t > 0$) and summing residues of $e^{st} F(s)$:

$$f(t) = \sum_{\text{poles } s_k} \text{Res}(e^{st} F(s), s = s_k), \quad (t > 0), \quad (97)$$

provided the closing arc contribution vanishes.

Key point. Laplace inversion is essentially a complex Fourier inversion with $s = \sigma + i\omega$, and the Bromwich contour integral is the rigorous inversion formula.

20 High-yield summary

- Definition:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t) dt.$$

- Derivatives:

$$\begin{aligned}\mathcal{L}\{f'\} &= sF - f(0), & \mathcal{L}\{f''\} &= s^2F - sf(0) - f'(0), \\ \mathcal{L}\{f^{(n)}\} &= s^n F - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0).\end{aligned}$$

- Delta and Heaviside:

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}, \quad \mathcal{L}\{H(t-a)\} = \frac{e^{-as}}{s}.$$

- Convolution:

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau) d\tau, \quad \mathcal{L}\{f * g\} = FG.$$

- Bromwich inversion:

$$f(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{st} F(s) ds.$$

21 Integral equations

21.1 General form

An integral equation is an equation where the unknown function $\phi(x)$ appears under an integral sign. A standard linear form is

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t) \phi(t) dt, \quad (98)$$

where

- $f(x)$ is known (driving term),
- $K(x, t)$ is the kernel,
- λ is a constant parameter,
- $\phi(x)$ is the unknown.

22 Types of integral equations

22.1 Fredholm vs Volterra

Fredholm. Limits are constants:

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t) \phi(t) dt. \quad (99)$$

Volterra. One limit depends on x (typical causal structure):

$$\phi(x) = f(x) + \lambda \int_a^x K(x, t) \phi(t) dt. \quad (100)$$

22.2 First kind vs second kind

First kind. Unknown appears only inside the integral:

$$f(x) = \int_a^b K(x, t) \phi(t) dt. \quad (101)$$

These are often ill-conditioned / non-unique.

Second kind. Unknown appears both outside and inside:

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t) \phi(t) dt. \quad (102)$$

This is typically better posed and supports iterative solution methods.

23 Neumann series (successive approximations)

23.1 Setup (Fredholm second kind)

Consider the Fredholm equation of the second kind

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t) \phi(t) dt. \quad (103)$$

Neumann's idea: iterate the equation by substituting approximations into the integral. A natural first guess is

$$\phi_0(x) = f(x). \quad (104)$$

Then define recursively

$$\phi_{n+1}(x) = f(x) + \lambda \int_a^b K(x, t) \phi_n(t) dt. \quad (105)$$

23.2 Series form

Iterating generates a power series in λ :

$$\phi(x) = \sum_{i=0}^{\infty} \lambda^i u_i(x), \quad (106)$$

with

$$u_0(x) = f(x), \quad (107)$$

$$u_1(x) = \int_a^b K(x, t_1) f(t_1) dt_1, \quad (108)$$

$$u_2(x) = \int_a^b \int_a^b K(x, t_1) K(t_1, t_2) f(t_2) dt_2 dt_1, \quad (109)$$

$$u_n(x) = \int_a^b \cdots \int_a^b K(x, t_1) K(t_1, t_2) \cdots K(t_{n-1}, t_n) f(t_n) dt_n \cdots dt_1. \quad (110)$$

This is exactly the Neumann series structure developed in Arfken & Weber. :contentReference[oaicite:2]index=2

23.3 Kernel iterates (iterated kernels)

Define the iterated kernels

$$K_1(x, t) = K(x, t), \quad (111)$$

$$K_{n+1}(x, t) = \int_a^b K(x, z) K_n(z, t) dz. \quad (112)$$

Then the Neumann series may be written compactly as

$$\phi(x) = f(x) + \lambda \int_a^b K_1(x, t) f(t) dt + \lambda^2 \int_a^b K_2(x, t) f(t) dt + \cdots. \quad (113)$$

23.4 Convergence idea

The Neumann series converges if the integral operator is “small enough” (in operator norm), or more precisely (as emphasized in the text) convergence is controlled by the eigenvalues of the corresponding homogeneous equation. :contentReference[oaicite:3]index=3

23.5 Volterra case (important simplification)

For Volterra equations of the second kind,

$$\phi(x) = f(x) + \lambda \int_a^x K(x, t) \phi(t) dt, \quad (114)$$

the Neumann series converges for all λ provided K is square integrable. :contentReference[oaicite:4]index=4

24 Separable (degenerate) kernels

24.1 Definition

A kernel is **separable** (degenerate) if it can be written as a finite sum

$$K(x, t) = \sum_{j=1}^n M_j(x) N_j(t), \quad (115)$$

as in A&W. :contentReference[oaicite:5]index=5 Example:

$$\cos(t - x) = \cos t \cos x + \sin t \sin x. \quad (116)$$

24.2 Fredholm second kind with separable kernel

Substitute into

$$\phi(x) = f(x) + \lambda \int_a^b K(x, t) \phi(t) dt \quad (117)$$

to obtain

$$\phi(x) = f(x) + \lambda \sum_{j=1}^n M_j(x) \int_a^b N_j(t) \phi(t) dt. \quad (118)$$

Define constants

$$c_j := \int_a^b N_j(t) \phi(t) dt. \quad (119)$$

Then the solution has the form

$$\phi(x) = f(x) + \lambda \sum_{j=1}^n c_j M_j(x). \quad (120)$$

24.3 Reduction to linear algebra

Multiply by $N_i(x)$ and integrate to obtain a finite linear system:

$$c_i = b_i + \lambda \sum_{j=1}^n a_{ij} c_j, \quad (121)$$

where

$$b_i = \int_a^b N_i(x) f(x) dx, \quad a_{ij} = \int_a^b N_i(x) M_j(x) dx. \quad (122)$$

Matrix form:

$$b = (1 - \lambda A)c, \quad c = (1 - \lambda A)^{-1}b. \quad (123)$$

This makes explicit the correspondence between integral equations and matrix equations. :contentReference[oaicite:6]index=6

24.4 Homogeneous case and eigenvalues

For the homogeneous equation $f(x) = 0$, we have $b = 0$ and

$$(1 - \lambda A)c = 0. \quad (124)$$

Nontrivial solutions exist only if

$$\det(1 - \lambda A) = 0. \quad (125)$$

This is the **secular equation** giving eigenvalues λ . :contentReference[oaicite:7]index=7

25 Homogeneous Fredholm equation of the second kind

25.1 Eigenvalue formulation

The homogeneous Fredholm equation of the second kind is

$$\phi(x) = \lambda \int_a^b K(x, t) \phi(t) dt. \quad (126)$$

A nontrivial solution $\phi(x) \neq 0$ exists only for special values $\lambda = \lambda_n$. These are the **eigenvalues**, and the corresponding solutions $\phi_n(x)$ are the **eigenfunctions**. :contentReference[oaicite:8]index=8

25.2 Symmetric kernels and Hilbert–Schmidt theory

Assume

$$K(x, t) = K(t, x), \quad K \in \mathbb{R}. \quad (127)$$

Then (Hilbert–Schmidt theory):

- eigenvalues λ_n are real,
- eigenfunctions corresponding to distinct eigenvalues are orthogonal,
- eigenfunctions form a complete set in a suitable sense.

This parallels Hermitian matrices and Sturm–Liouville ODEs. :contentReference[oaicite:9]index=9

25.3 Orthogonality of eigenfunctions

Let ϕ_i, ϕ_j satisfy

$$\phi_i(x) = \lambda_i \int_a^b K(x, t) \phi_i(t) dt, \quad \phi_j(x) = \lambda_j \int_a^b K(x, t) \phi_j(t) dt.$$

For a symmetric kernel, one obtains

$$(\lambda_j - \lambda_i) \int_a^b \phi_i(x) \phi_j(x) dx = 0. \quad (128)$$

Thus if $\lambda_i \neq \lambda_j$,

$$\boxed{\int_a^b \phi_i(x) \phi_j(x) dx = 0.} \quad (129)$$

This is the Fredholm integral-equation analog of Sturm–Liouville orthogonality. :contentReference[oaicite:10]index=10

25.4 Normalization and degeneracy

One may normalize eigenfunctions so that

$$\int_a^b \phi_i(x) \phi_j(x) dx = \delta_{ij}. \quad (130)$$

If an eigenvalue is degenerate, the corresponding eigenfunctions may be orthogonalized by Gram–Schmidt. :contentReference[oaicite:11]index=11

25.5 Reality of eigenvalues

For Hermitian kernels $K(x, t) = K^*(t, x)$ (general complex case), eigenvalues are real. :contentReference[oaicite:12]index=12

26 Kernel expansion in eigenfunctions (bilinear expansion)

For symmetric kernels, the kernel itself can be expanded in eigenfunctions:

$$\boxed{K(x, t) = \sum_{n=1}^{\infty} \frac{\phi_n(x) \phi_n(t)}{\lambda_n}, \quad (0 \text{ not an eigenvalue}).} \quad (131)$$

This is the bilinear eigenfunction expansion stated in A&W. :contentReference[oaicite:13]index=13

27 High-yield summary

- Fredholm: fixed limits; Volterra: variable limit.
- First kind: $f = \int K \phi$ (often non-unique/ill-conditioned).

- Second kind: $\phi = f + \lambda \int K \phi$ (iterative methods work).
- Neumann series: iterate substitution to get a power series in λ :

$$\phi = \sum_{n=0}^{\infty} \lambda^n u_n.$$

- Separable (degenerate) kernels reduce the integral equation to a finite matrix equation:

$$(1 - \lambda A)c = b.$$

- Homogeneous Fredholm II: $\phi = \lambda \int K \phi$ has nontrivial solutions only for eigenvalues λ_n .
- For symmetric kernels: eigenvalues real, eigenfunctions orthogonal, completeness holds.
- Kernel eigenfunction expansion:

$$K(x, t) = \sum_n \frac{\phi_n(x)\phi_n(t)}{\lambda_n}.$$

28 Variational calculus (calculus of variations)

28.1 Basic variational problem

A standard variational problem is to find a function $y(x)$ that extremizes (functional is stationary):

$$J[y] = \int_{x_1}^{x_2} F(x, y(x), y'(x)) dx. \quad (132)$$

We consider variations

$$y(x) \rightarrow y(x) + \epsilon \eta(x), \quad (133)$$

where $\eta(x)$ is an arbitrary smooth test function satisfying boundary conditions.

28.2 Fixed-endpoint conditions

If endpoints are fixed:

$$y(x_1) = y_1, \quad y(x_2) = y_2, \quad (134)$$

then the variation must satisfy

$$\eta(x_1) = \eta(x_2) = 0. \quad (135)$$

28.3 First variation and stationarity

The condition for an extremum is

$$\delta J = 0 \quad \text{for all admissible } \eta(x). \quad (136)$$

Compute:

$$\delta J = \left. \frac{d}{d\epsilon} J[y + \epsilon \eta] \right|_{\epsilon=0} \quad (137)$$

$$= \int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) dx. \quad (138)$$

Integrate by parts the η' term:

$$\int_{x_1}^{x_2} \frac{\partial F}{\partial y'} \eta' dx = \left[\frac{\partial F}{\partial y'} \eta \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \eta dx. \quad (139)$$

For fixed endpoints, the boundary term vanishes, so

$$\delta J = \int_{x_1}^{x_2} \left(\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} \right) \eta dx. \quad (140)$$

Since $\eta(x)$ is arbitrary, the integrand must vanish.

29 Euler equation (Euler–Lagrange equation)

29.1 Euler–Lagrange equation (1 function)

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.} \quad (141)$$

29.2 Multiple dependent variables

If $y_i(x)$, $i = 1, \dots, n$:

$$J[\{y_i\}] = \int_{x_1}^{x_2} F(x, y_1, \dots, y_n, y'_1, \dots, y'_n) dx, \quad (142)$$

then for each i :

$$\boxed{\frac{\partial F}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'_i} \right) = 0.} \quad (143)$$

29.3 Special case: F does not depend explicitly on y

If $\partial F / \partial y = 0$, then

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \Rightarrow \quad \frac{\partial F}{\partial y'} = \text{constant}. \quad (144)$$

29.4 Special case: F does not depend explicitly on x (first integral)

If $\partial F / \partial x = 0$, then

$$\boxed{F - y' \frac{\partial F}{\partial y'} = \text{constant}.} \quad (145)$$

(This is the variational analog of energy conservation.)

30 Variational problems with constraints

Constraints restrict the admissible functions $y(x)$.

30.1 Integral constraint (isoperimetric problem)

Suppose we extremize $J[y]$ subject to

$$I[y] = \int_{x_1}^{x_2} G(x, y, y') dx = C, \quad (146)$$

where C is fixed.

30.2 Method of Lagrange multipliers (functional form)

Introduce a constant multiplier λ and define the augmented functional

$$\tilde{J}[y] = \int_{x_1}^{x_2} \left(F(x, y, y') + \lambda G(x, y, y') \right) dx. \quad (147)$$

Then the constrained extremum is obtained by solving

$$\delta \tilde{J} = 0 \quad (148)$$

together with the constraint $I[y] = C$.

Thus the Euler–Lagrange equation becomes

$$\boxed{\frac{\partial}{\partial y}(F + \lambda G) - \frac{d}{dx} \left(\frac{\partial}{\partial y'}(F + \lambda G) \right) = 0,} \quad (149)$$

with λ determined from the constraint.

30.3 Pointwise constraint

If there is a constraint that must hold at each x :

$$g(x, y, y') = 0, \quad (150)$$

then λ may need to be a function $\lambda(x)$ and we extremize

$$\tilde{J}[y] = \int_{x_1}^{x_2} \left(F(x, y, y') + \lambda(x) g(x, y, y') \right) dx. \quad (151)$$

Euler–Lagrange gives coupled equations for $y(x)$ and $\lambda(x)$:

$$\frac{\partial}{\partial y}(F + \lambda g) - \frac{d}{dx} \left(\frac{\partial}{\partial y'}(F + \lambda g) \right) = 0, \quad (152)$$

and the constraint $g(x, y, y') = 0$.

30.4 Multiple constraints

For m integral constraints

$$\int G_\alpha dx = C_\alpha, \quad \alpha = 1, \dots, m,$$

use

$$\tilde{J} = \int \left(F + \sum_{\alpha=1}^m \lambda_\alpha G_\alpha \right) dx. \quad (153)$$

31 Boundary conditions and transversality (quick)

31.1 Fixed endpoints

If $y(x_1), y(x_2)$ are fixed, boundary terms vanish automatically.

31.2 Free endpoint (natural boundary condition)

If $y(x_2)$ is free, then $\eta(x_2)$ is not forced to be zero. From the boundary term

$$\left[\frac{\partial F}{\partial y'} \eta \right]_{x_1}^{x_2},$$

we obtain the natural condition at x_2 :

$$\boxed{\left. \frac{\partial F}{\partial y'} \right|_{x=x_2} = 0 \quad (\text{if } y(x_2) \text{ free}).} \quad (154)$$

Similarly for x_1 .

32 High-yield summary

- Functional:

$$J[y] = \int_{x_1}^{x_2} F(x, y, y') dx.$$

- Variation: $y \rightarrow y + \epsilon \eta$, stationarity $\delta J = 0$.

- Euler–Lagrange:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

- If $\partial F / \partial x = 0$:

$$F - y' \frac{\partial F}{\partial y'} = \text{const.}$$

- Integral constraint $\int G dx = C$:

$$F \rightarrow F + \lambda G.$$

- Pointwise constraint $g(x, y, y') = 0$:

$$F \rightarrow F + \lambda(x)g.$$

33 Concept of nonlocality

33.1 Local vs nonlocal description

A **local** theory determines the evolution at (x, t) using only values at that point (and finitely many derivatives):

$$\mathcal{E}(u(x, t), \partial u(x, t), \partial^2 u(x, t), \dots) = 0. \quad (155)$$

A **nonlocal** theory allows the state at (x, t) to depend on values in a region of space and/or the history in time:

$$u(x, t) = \int_{\Omega} K(x, \xi) g(\xi, t) d\xi \quad (\text{spatial nonlocality}), \quad (156)$$

$$u(x, t) = \int_0^t M(t - \tau) g(x, \tau) d\tau \quad (\text{temporal nonlocality / memory}). \quad (157)$$

Key physical interpretation (Herrmann). Nonlocal effects may occur in **space and time**. In the time domain, nonlocality appears as a **memory effect**: the current behavior depends not only on the present state but also on the past. :contentReference[oaicite:1]index=1

34 Volterra principles of nonlocality

34.1 Volterra (hereditary) principle

Volterra's viewpoint: in systems with memory, the response at time t depends on the full past history through a **causal integral operator**:

$$R(t) = R_0(t) + \int_0^t K(t, \tau) I(\tau) d\tau, \quad (t \geq 0). \quad (158)$$

Here $I(\tau)$ is the “input” (excitation/forcing), $R(t)$ the response, and K is a memory kernel.

34.2 Volterra superposition principle (linear memory)

If the system is linear, contributions from different past times add:

$$R(t) = \int_0^t M(t - \tau) I(\tau) d\tau, \quad (159)$$

i.e. a **Volterra convolution** with a kernel M depending only on time difference.

34.3 Causality constraint

The Volterra form enforces causality automatically:

$$\tau > t \Rightarrow \text{no influence on } R(t), \quad \text{hence integration domain } [0, t]. \quad (160)$$

34.4 Geometric memory effect (Herrmann example)

Herrmann gives a simple geometric mechanism: adding boundaries to an otherwise free system can produce a delayed contribution to the source term, e.g.

$$\psi(t) \longrightarrow \psi(t) + \rho \psi(t - \tau), \quad (161)$$

where the second term is a delayed (nonlocal-in-time) contribution generated by reflections. :contentReference[oaicite:2]index=2

Important point. The system does not “store intelligence”; memory is an emergent effect from the dynamics (e.g. reflections/boundaries) rather than literal memorization. :contentReference[oaicite:3]index=3

35 Nonlocal operations (nonlocal operators)

35.1 Spatial nonlocal operator (integral kernel)

A general spatial nonlocal operator is

$$(\mathcal{K}u)(x) = \int_{\Omega} K(x, \xi) u(\xi) d\xi. \quad (162)$$

This contrasts with local operators like $\partial_x u(x)$ which only use infinitesimal neighborhood information.

35.2 Translation-invariant kernels $\Rightarrow$ convolution

If $K(x, \xi) = K(x - \xi)$, then

$$(\mathcal{K}u)(x) = (K * u)(x) = \int_{\mathbb{R}} K(x - \xi) u(\xi) d\xi. \quad (163)$$

35.3 Temporal nonlocal operator (memory kernel)

A general causal memory operator is

$$(\mathcal{M}u)(t) = \int_0^t M(t - \tau) u(\tau) d\tau. \quad (164)$$

35.4 Kernel types (how memory/range behaves)

- **Short memory:** $M(t) \sim e^{-t/\tau}$ (finite relaxation scale).
- **Long memory:** $M(t) \sim t^{-\alpha}$ (power-law tail; history never fully dies).
- **Local limit:** $M(t) \rightarrow \delta(t)$ gives $(\mathcal{M}u)(t) = u(t)$.

36 Relation to fractional calculus

36.1 Fractional calculus as a subclass of nonlocal theories

Herrmann emphasizes that an important subgroup of nonlocal field theories can be described by **fractional operators**, i.e. derivatives of non-integer order:

$$\frac{d^n}{dx^n} \longrightarrow \frac{d^\alpha}{dx^\alpha}, \quad \alpha \in \mathbb{R}, \mathbb{C} \text{ (or even } \alpha = \alpha(x, t) \text{)}. \quad :contentReference[oaicite:4]index=4 \quad (165)$$

36.2 Why fractional operators are nonlocal

Fractional derivatives typically involve integrals with weakly singular kernels. This means the derivative at t depends on values for all $\tau < t$ (memory).

36.3 Fractional integral (Volterra operator)

For $\alpha > 0$, the Riemann–Liouville fractional integral is

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau. \quad (166)$$

This is exactly a Volterra convolution:

$$(I^\alpha f)(t) = (M_\alpha * f)(t), \quad M_\alpha(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}.$$

36.4 Fractional derivatives (Caputo vs Riemann–Liouville)

For $0 < \alpha < 1$:

Riemann–Liouville derivative.

$$(D_t^\alpha f)(t) = \frac{d}{dt} \left[\frac{1}{\Gamma(1-\alpha)} \int_0^t (t - \tau)^{-\alpha} f(\tau) d\tau \right]. \quad (167)$$

Caputo derivative.

$$({}^C D_t^\alpha f)(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t - \tau)^{-\alpha} f'(\tau) d\tau. \quad (168)$$

Difference (important in physics). Caputo gives ${}^C D^\alpha(\text{const}) = 0$ while Riemann–Liouville does not. Herrmann explicitly highlights this as a motivation for Caputo-type definitions. :contentReference[oaicite:5]index=5

36.5 Fractional derivative defined via eigenfunctions (Herrmann’s “historical” route)

Herrmann presents an intuitive definition based on preserving the integer-order derivative rules for special function classes (then extending $n \rightarrow \alpha$). For example: [:contentReference\[oaicite:6\]index=6](#)

$$\frac{d^\alpha}{dx^\alpha} e^{kx} = k^\alpha e^{kx}, \quad k \geq 0, \quad (169)$$

$$\frac{d^\alpha}{dx^\alpha} \sin(kx) = k^\alpha \sin\left(kx + \frac{\pi\alpha}{2}\right), \quad k \geq 0, \quad (170)$$

$$\frac{d^\alpha}{dx^\alpha} x^k = \frac{\Gamma(1+k)}{\Gamma(1+k-\alpha)} x^{k-\alpha}, \quad x \geq 0. \quad (171)$$

36.6 Nonlocality via infinite-derivative expansion

Herrmann also shows that fractional differentiation can be written as an infinite series in integer derivatives (binomial-type expansion):

$$\left(\frac{d}{dx} + \omega\right)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} \omega^{\alpha-k} \frac{d^k}{dx^k}. \quad (172)$$

Since infinitely many derivatives are required, one needs essentially the full Taylor data of f , which is another way to see **nonlocality** (global information requirement). [:contentReference\[oaicite:7\]index=7](#)

36.7 Memory kernels $\Rightarrow$ fractional dynamics (core bridge)

Volterra memory:

$$R(t) = \int_0^t M(t-\tau) I(\tau) d\tau$$

becomes fractional when the kernel has a power-law form:

$$M(t) \propto t^{-\alpha} \quad \Rightarrow \quad R(t) \propto {}^C D_t^\alpha I(t) \text{ or } I^\alpha I(t), \quad (173)$$

depending on the constitutive structure.

Interpretation.

- Exponential kernels $\Rightarrow$ short memory, Markov-like relaxation.
- Power-law kernels $\Rightarrow$ long memory, fractional operators.

37 High-yield summary

- **Nonlocality** = dependence on a region in space or the entire past in time.
- **Volterra nonlocality** = causal memory integral on $[0, t]$:

$$R(t) = \int_0^t M(t-\tau) I(\tau) d\tau.$$

- **Nonlocal operators** are integral-kernel operators in space/time.
- **Fractional calculus** is a structured way to build nonlocal operators:

$$\frac{d^n}{dx^n} \rightarrow \frac{d^\alpha}{dx^\alpha}.$$

- **Fractional integrals/derivatives** are nonlocal because they use Volterra-type power-law kernels:

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau.$$

- Herrmann's key message: fractional operators provide a practical framework for nonlocal theories with memory effects. :contentReference[oaicite:8]index=8

38 Random walks and diffusion processes

38.1 Discrete-time random walk (1D)

Let X_n be the position after n steps:

$$X_n = \sum_{k=1}^n \xi_k, \quad \xi_k \in \{+a, -a\}, \quad \mathbb{P}(\xi_k = +a) = p, \quad \mathbb{P}(\xi_k = -a) = q = 1 - p. \quad (174)$$

Mean and variance:

$$\mathbb{E}[X_n] = na(p - q), \quad \text{Var}(X_n) = 4npq a^2. \quad (175)$$

Symmetric walk ($p = q = 1/2$):

$$\mathbb{E}[X_n] = 0, \quad \text{Var}(X_n) = na^2. \quad (176)$$

38.2 Central limit theorem (Gaussian scaling)

If ξ_k are i.i.d. with finite variance σ^2 , then for large n :

$$\frac{X_n - n\mu}{\sqrt{n\sigma^2}} \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, 1), \quad \mu = \mathbb{E}[\xi_k]. \quad (177)$$

So the displacement distribution approaches a Gaussian.

39 Normal diffusion

39.1 Mean squared displacement (MSD)

For a stochastic process $X(t)$ define

$$\langle X^2(t) \rangle := \mathbb{E}[X(t)^2]. \quad (178)$$

Normal diffusion is characterized by linear growth:

$$\langle X^2(t) \rangle \sim 2Dt \quad (1D), \quad \langle r^2(t) \rangle \sim 2dDt \quad (\text{in } d \text{ dimensions}). \quad (179)$$

39.2 Diffusion equation (Fick / heat equation)

Let $P(x, t)$ be the probability density of being at x at time t . Normal diffusion satisfies:

$$\frac{\partial P}{\partial t} = D \frac{\partial^2 P}{\partial x^2}. \quad (180)$$

With initial condition $P(x, 0) = \delta(x)$, the Green function is Gaussian:

$$P(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right). \quad (181)$$

40 Anomalous diffusion

40.1 Definition via MSD scaling

Anomalous diffusion is defined by a power-law MSD:

$$\langle X^2(t) \rangle \sim t^\alpha, \quad \alpha \neq 1. \quad (182)$$

Classification:

- $0 < \alpha < 1$: **subdiffusion** (slower than normal).
- $\alpha = 1$: **normal diffusion**.
- $1 < \alpha < 2$: **superdiffusion**.
- $\alpha = 2$: **ballistic transport** ($X(t) \sim vt$).

40.2 Physical origins

Common mechanisms:

- **Trapping / waiting times** (long rests) $\Rightarrow$ subdiffusion.
- **Long jumps / flights** (rare huge displacements) $\Rightarrow$ superdiffusion.
- **Long-time correlations** (persistent velocity) $\Rightarrow$ superdiffusion.
- **Fractal / disordered media** $\Rightarrow$ anomalous scaling.

40.3 Connection to fractional calculus (Herrmann motivation)

Herrmann highlights that anomalous diffusion and fractional Brownian motion became a major physics motivation for fractional methods (especially since the 1980s). :contentReference[oaicite:2]index=2

Core idea:

- **Temporal nonlocality / memory** $\Rightarrow$ fractional *time* derivatives.
- **Spatial nonlocality / long jumps** $\Rightarrow$ fractional *space* derivatives.

41 Continuous-time random walk (CTRW) framework

41.1 CTRW definition

A CTRW consists of:

- jump lengths $\eta_1, \eta_2, \dots$ with PDF $\lambda(\eta)$
- waiting times $\tau_1, \tau_2, \dots$ with PDF $\psi(\tau)$

and position after $N(t)$ jumps:

$$X(t) = \sum_{k=1}^{N(t)} \eta_k, \quad N(t) = \max \left\{ n : \sum_{k=1}^n \tau_k \leq t \right\}. \quad (183)$$

41.2 Normal diffusion condition

If $\mathbb{E}[\tau] < \infty$ and $\mathbb{E}[\eta^2] < \infty$, then at large times:

$$\langle X^2(t) \rangle \sim t.$$

41.3 Subdiffusion from heavy-tailed waiting times

If the waiting time distribution has a power-law tail:

$$\psi(\tau) \sim \tau^{-1-\beta}, \quad 0 < \beta < 1, \quad (184)$$

then $\mathbb{E}[\tau] = \infty$ and typically

$$\langle X^2(t) \rangle \sim t^\beta \quad (\text{subdiffusion}). \quad (185)$$

This is a standard route to fractional-in-time diffusion equations.

41.4 Superdiffusion from heavy-tailed jump lengths

If jump lengths have heavy tails:

$$\lambda(\eta) \sim |\eta|^{-1-\mu}, \quad 0 < \mu < 2, \quad (186)$$

then $\mathbb{E}[\eta^2] = \infty$ and the process can become superdiffusive. This is associated with Lévy stable statistics and fractional Laplacians.

42 Lévy flights vs Lévy walks

42.1 Lévy flight (mathematical idealization)

A **Lévy flight** is a random walk with i.i.d. heavy-tailed step lengths:

$$\lambda(\eta) \sim |\eta|^{-1-\mu}, \quad 0 < \mu < 2. \quad (187)$$

Key property: the variance diverges:

$$\mathbb{E}[\eta^2] = \infty. \quad (188)$$

Problem: steps occur in effectively zero time, so arbitrarily long jumps happen instantly (unphysical for finite-speed transport).

42.2 Lévy walk (physically meaningful)

A **Lévy walk** couples displacement and travel time. A typical model:

$$\eta = v \tau \cdot s, \quad s \in \{+1, -1\}, \quad (189)$$

where τ is a flight time and v is a finite speed.

If flight times are heavy-tailed:

$$\psi(\tau) \sim \tau^{-1-\gamma}, \quad (190)$$

then transport can be superdiffusive but still respects finite propagation speed.

42.3 Scaling behavior (qualitative)

Depending on γ :

- large γ (short flights dominate) $\Rightarrow$ normal diffusion
- intermediate $\gamma \Rightarrow$ superdiffusion
- very small $\gamma \Rightarrow$ nearly ballistic scaling

42.4 Why Lévy walks matter

Lévy walks provide:

- heavy tails (rare long moves)
- finite velocity constraint
- realistic superdiffusion model in physics/biology

43 Fractional diffusion equations (bridge to anomalous diffusion)

43.1 Fractional time diffusion (subdiffusion)

Subdiffusion with memory can be modeled by a fractional time derivative:

$${}^C D_t^\beta P(x, t) = D \frac{\partial^2 P}{\partial x^2}, \quad 0 < \beta < 1. \quad (191)$$

This encodes **nonlocality in time** (history dependence).

43.2 Fractional space diffusion (superdiffusion)

Superdiffusion with long jumps can be modeled using a fractional Laplacian:

$$\frac{\partial P}{\partial t} = -K_\mu (-\Delta)^{\mu/2} P, \quad 0 < \mu < 2. \quad (192)$$

This encodes **nonlocality in space** (long-range jumps).

43.3 Combined space-time fractional transport

More general anomalous transport can involve both:

$${}^C D_t^\beta P(x, t) = -K_\mu (-\Delta)^{\mu/2} P. \quad (193)$$

44 High-yield summary

- **Random walk** $\Rightarrow$ diffusion in the continuum limit.
- **Normal diffusion:** $\langle X^2(t) \rangle \sim t$ and Gaussian $P(x, t)$.
- **Anomalous diffusion:** $\langle X^2(t) \rangle \sim t^\alpha$, $\alpha \neq 1$.
- **Subdiffusion** comes from long waiting times / trapping $\Rightarrow$ memory.
- **Superdiffusion** comes from long jumps / persistent motion.
- **Lévy flight:** heavy-tailed jumps, infinite variance, infinite-speed issue.
- **Lévy walk:** heavy-tailed *flight times* with finite velocity (physical).
- **Fractional calculus link (Herrmann):** anomalous diffusion motivates fractional operators as nonlocal descriptions of transport. :contentReference[oaicite:3]index=3

45 Concept of self-similarity

45.1 Exact self-similarity (scale invariance)

A function/object is **self-similar** if it reproduces its form under rescaling.

Functional form. A function $f(x)$ is scale invariant if

$$f(\lambda x) = \lambda^k f(x) \quad \text{for all } \lambda > 0, \quad (194)$$

for some exponent $k \in \mathbb{R}$.

Solution. The only (nontrivial) continuous solutions are power laws:

$$f(x) = C x^k. \quad (195)$$

45.2 Self-similarity in stochastic processes

A stochastic process $X(t)$ is **self-similar** with parameter H (Hurst exponent) if

$$X(\lambda t) \stackrel{d}{=} \lambda^H X(t), \quad \lambda > 0, \quad (196)$$

where $\stackrel{d}{=}$ means equality in distribution.

Typical consequences:

$$\mathbb{E}[X(t)] \sim t^H, \quad \mathbb{E}[X^2(t)] \sim t^{2H}. \quad (197)$$

Diffusion scaling link. If MSD scales as

$$\langle X^2(t) \rangle \sim t^\alpha,$$

then typically

$$\alpha = 2H. \quad (198)$$

46 Power-law functions

46.1 Definition

A **power law** is any relation of the form

$$y(x) = C x^k, \quad x > 0. \quad (199)$$

46.2 Key properties

- **Scale invariance:**

$$y(\lambda x) = C(\lambda x)^k = \lambda^k y(x).$$

- **Straight line on log-log plot:**

$$\log y = \log C + k \log x. \quad (200)$$

So slope in a $\log y$ vs $\log x$ plot equals k .

- **No intrinsic scale:** power laws do not define a characteristic length/time.

46.3 Power-law tails (heavy tails)

A probability density can have a power-law tail:

$$p(x) \sim \frac{1}{|x|^{1+\mu}} \quad (|x| \rightarrow \infty). \quad (201)$$

Moments may diverge:

$$\mathbb{E}[|X|^q] < \infty \iff q < \mu. \quad (202)$$

Example: if $0 < \mu < 2$, then variance diverges.

46.4 Why power laws appear

Power laws typically arise from:

- scale invariance / self-similarity
- multiplicative processes
- critical phenomena (no characteristic scale)
- fractal geometry
- long-memory kernels (e.g. $t^{-\alpha}$)

47 Fractals

47.1 Basic idea

A **fractal** is a set/geometry with:

- structure at many scales (self-similarity),
- non-integer (fractional) dimension,
- often power-law scaling of measured quantities.

47.2 Self-similar fractals and fractal dimension

Suppose a set is made of N self-similar copies of itself, each scaled by a factor r (with $0 < r < 1$).

Then the **similarity dimension** D satisfies

$$N = r^{-D} \iff D = \frac{\log N}{\log(1/r)}. \quad (203)$$

Examples.

- **Cantor set:** $N = 2, r = 1/3$

$$D = \frac{\log 2}{\log 3} \approx 0.6309.$$

- **Koch curve:** $N = 4, r = 1/3$

$$D = \frac{\log 4}{\log 3} \approx 1.2619.$$

- **Sierpiński triangle:** $N = 3, r = 1/2$

$$D = \frac{\log 3}{\log 2} \approx 1.585.$$

47.3 Box-counting (Minkowski) dimension

Cover the set S with boxes of side length ε . Let $N(\varepsilon)$ be the minimal number needed.

The **box-counting dimension** is

$$D_B = \lim_{\varepsilon \rightarrow 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}, \quad (204)$$

if the limit exists.

47.4 Scaling laws on fractals

On fractals, measured quantities often scale with non-integer exponents.

Example: “mass” or number of points inside radius R :

$$M(R) \sim R^D, \quad (205)$$

where D is the fractal dimension.

47.5 Random walks on fractals (important physics link)

Diffusion on fractal/disordered structures often becomes anomalous:

$$\langle r^2(t) \rangle \sim t^\alpha, \quad \alpha \neq 1. \quad (206)$$

This is one route to **fractional** models of transport.

48 Connection to fractional calculus (Herrmann motivation)

Herrmann notes that Mandelbrot’s fractal geometry (1980s) strongly attracted physicists and produced major interest in fractional Brownian motion and anomalous diffusion, which are key motivations for fractional calculus methods in physics. :contentReference[oaicite:1]index=1

48.1 Why fractals naturally lead to fractional operators

Heuristic links:

- **Fractal geometry** $\Rightarrow$ effective non-integer dimension / scaling.
- **Power-law memory kernels** $\Rightarrow$ nonlocal time operators.
- **Long-range jump statistics** $\Rightarrow$ fractional Laplacians in space.

Thus, fractional derivatives are natural mathematical tools to encode **scale-free** and **nonlocal** behavior.

49 High-yield summary

- **Self-similarity** = invariance under scaling.
- **Scale invariance equation:** $f(\lambda x) = \lambda^k f(x)$.
- **Power laws** are the unique continuous solutions: $f(x) = Cx^k$.
- **Power laws** appear as straight lines on log–log plots.
- **Fractals** = self-similar sets with non-integer dimension.
- **Similarity dimension:** $D = \log N / \log(1/r)$.
- **Box-counting dimension:** $D_B = \lim_{\varepsilon \rightarrow 0} \log N(\varepsilon) / \log(1/\varepsilon)$.
- **Physics link:** fractals $\Rightarrow$ anomalous diffusion $\Rightarrow$ fractional calculus models. :contentReference[oaicite:2]index=2

50 Fractional calculus

50.1 Core idea

Fractional calculus extends differentiation/integration from integer order $n \in \mathbb{N}$ to non-integer order $\alpha \in \mathbb{R}$ (and sometimes complex α):

$$\frac{d^n}{dx^n} \longrightarrow \frac{d^\alpha}{dx^\alpha}. \quad (207)$$

This provides a systematic framework for **nonlocal operators** and **power-law memory** in physics.
:contentReference[oaicite:2]index=2

50.2 Gamma function (replaces factorials)

Fractional calculus heavily uses the Gamma function:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad \Gamma(n+1) = n!. \quad (208)$$

51 Fractional integrals (Riemann–Liouville)

51.1 Riemann–Liouville fractional integral

For $\alpha > 0$, define the fractional integral:

$$(I_{0+}^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-\tau)^{\alpha-1} f(\tau) d\tau. \quad (209)$$

This is a Volterra convolution with a power-law kernel, hence it is **nonlocal in time**.

51.2 Semigroup property

For suitable f ,

$$I^\alpha I^\beta f = I^{\alpha+\beta} f. \quad (210)$$

52 Fractional derivatives: main definitions

There is no unique definition of a fractional derivative. Different definitions agree on broad classes of functions but differ in:

- how initial conditions are handled,
- whether constants differentiate to zero,
- regularity requirements.

52.1 Riemann–Liouville (RL) fractional derivative

For $n-1 < \alpha < n$ with $n \in \mathbb{N}$:

$$(D_{0+}^\alpha f)(t) = \frac{d^n}{dt^n} (I_{0+}^{n-\alpha} f)(t) = \frac{1}{\Gamma(n-\alpha)} \frac{d^n}{dt^n} \int_0^t (t-\tau)^{n-\alpha-1} f(\tau) d\tau. \quad (211)$$

Key feature. RL derivative is naturally tied to fractional integrals and is mathematically clean, but it gives (typically)

$$D_{0+}^{\alpha}(1) \neq 0. \quad (212)$$

This can be inconvenient in physics initial-value problems.

52.2 Caputo fractional derivative

For $n - 1 < \alpha < n$:

$$\boxed{({}^C D_{0+}^{\alpha} f)(t) = I_{0+}^{n-\alpha} \left(\frac{d^n f}{dt^n} \right) (t) = \frac{1}{\Gamma(n-\alpha)} \int_0^t (t-\tau)^{n-\alpha-1} f^{(n)}(\tau) d\tau.} \quad (213)$$

For $0 < \alpha < 1$ this simplifies to

$${}^C D_{0+}^{\alpha} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-\tau)^{-\alpha} f'(\tau) d\tau. \quad (214)$$

Key feature (physics advantage). Caputo derivative satisfies

$${}^C D_{0+}^{\alpha}(1) = 0, \quad (215)$$

and initial conditions appear in the usual form $f(0), f'(0), \dots$. Herrmann emphasizes this as a major motivation for Caputo-type derivatives. :contentReference[oaicite:3]index=3

52.3 Grünwald–Letnikov (GL) fractional derivative

A definition closer to the discrete difference quotient:

$$\boxed{(D^{\alpha} f)(t) = \lim_{h \rightarrow 0^+} \frac{1}{h^{\alpha}} \sum_{k=0}^{\lfloor t/h \rfloor} (-1)^k \binom{\alpha}{k} f(t - kh),} \quad (216)$$

where the generalized binomial coefficient is

$$\binom{\alpha}{k} = \frac{\Gamma(\alpha+1)}{\Gamma(k+1)\Gamma(\alpha-k+1)}. \quad (217)$$

Use. GL is useful for numerical discretizations and shows explicitly the **memory** structure: $f(t)$ depends on all past values $f(t - kh)$.

52.4 Fourier/Laplace definitions (spectral characterization)

Fractional derivatives can be defined via transforms.

Fourier-space (spatial derivative). If $\mathcal{F}\{f(x)\}(k) = \tilde{f}(k)$, define

$$\mathcal{F} \left\{ \frac{d^{\alpha} f}{dx^{\alpha}} \right\} (k) = (ik)^{\alpha} \tilde{f}(k), \quad (218)$$

with a suitable branch choice for $(ik)^{\alpha}$.

Laplace-space (time derivative). For Caputo derivative ($0 < \alpha < 1$),

$$\boxed{\mathcal{L}\{ {}^C D_t^\alpha f(t) \}(s) = s^\alpha F(s) - s^{\alpha-1} f(0).} \quad (219)$$

This is one of the most useful formulas in solving fractional ODEs.

53 Herrmann’s “historical” motivation / consistency conditions

Herrmann motivates fractional derivatives by demanding that standard derivative rules extend smoothly to non-integer order for simple eigenfunctions. :contentReference[oaicite:4]index=4

53.1 Exponentials as eigenfunctions

$$\boxed{\frac{d^\alpha}{dx^\alpha} e^{kx} = k^\alpha e^{kx}, \quad k \geq 0.} \quad (220)$$

53.2 Trigonometric functions

$$\boxed{\frac{d^\alpha}{dx^\alpha} \sin(kx) = k^\alpha \sin\left(kx + \frac{\pi\alpha}{2}\right), \quad k \geq 0.} \quad (221)$$

53.3 Power functions

For $x \geq 0$,

$$\boxed{\frac{d^\alpha}{dx^\alpha} x^m = \frac{\Gamma(m+1)}{\Gamma(m+1-\alpha)} x^{m-\alpha}.} \quad (222)$$

These identities are consistent with integer derivatives when $\alpha = n \in \mathbb{N}$.

54 RL vs Caputo (quick comparison)

For $0 < \alpha < 1$:

- RL:

$$D^\alpha f(t) = \frac{d}{dt} \left[\frac{1}{\Gamma(1-\alpha)} \int_0^t (t-\tau)^{-\alpha} f(\tau) d\tau \right].$$

- Caputo:

$${}^C D^\alpha f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-\tau)^{-\alpha} f'(\tau) d\tau.$$

Relation (for sufficiently smooth f):

$${}^C D^\alpha f(t) = D^\alpha (f(t) - f(0)). \quad (223)$$

55 High-yield summary

- Fractional calculus extends derivatives/integrals to order $\alpha \notin \mathbb{N}$.
- RL fractional integral:

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau.$$

- RL derivative:

$$D^\alpha f = \frac{d^n}{dt^n} I^{n-\alpha} f, \quad n-1 < \alpha < n.$$

- Caputo derivative:

$${}^C D^\alpha f = I^{n-\alpha} f^{(n)}, \quad n-1 < \alpha < n,$$

and ${}^C D^\alpha(\text{const}) = 0$.

- GL derivative gives a discrete memory sum and is useful numerically.
- Transform rule:

$$\mathcal{L}\{{}^C D_t^\alpha f\} = s^\alpha F(s) - s^{\alpha-1} f(0).$$

- Herrmann motivation: preserve eigenfunction rules (e.g. $d^\alpha e^{kx}/dx^\alpha = k^\alpha e^{kx}$). :contentReference[oaicite:5]index=5

56 Fractional calculus: fractional integration

56.1 Motivation

Fractional integration generalizes the n -fold integral

$$I^n f(t) = \underbrace{\int_0^t \int_0^{\tau_1} \cdots \int_0^{\tau_{n-1}}}_{n \text{ times}} f(\tau_n) d\tau_n \cdots d\tau_1$$

to non-integer order $\alpha > 0$:

$$I^n \longrightarrow I^\alpha.$$

Fractional integrals are **nonlocal operators**: $(I^\alpha f)(t)$ depends on all values of $f(\tau)$ for $0 \leq \tau \leq t$.
:contentReference[oaicite:1]index=1

57 Riemann–Liouville (RL) fractional integral

57.1 Definition

For $\alpha > 0$, the (left-sided) RL fractional integral is

$$(I_{0+}^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau, \quad t > 0. \quad (224)$$

The kernel is a power law:

$$M_\alpha(t - \tau) = \frac{(t - \tau)^{\alpha-1}}{\Gamma(\alpha)}.$$

Thus I^α is a Volterra-type convolution operator.

57.2 Integer-order consistency

If $\alpha = n \in \mathbb{N}$, then

$$(I^n f)(t) = \frac{1}{(n-1)!} \int_0^t (t - \tau)^{n-1} f(\tau) d\tau, \quad (225)$$

which reproduces the standard n -fold integral.

57.3 Semigroup property

For suitable f ,

$$I^\alpha I^\beta f = I^{\alpha+\beta} f, \quad \alpha, \beta > 0. \quad (226)$$

57.4 Identity element

$$I^0 f = f. \quad (227)$$

58 Right-sided RL fractional integral

On an interval $[a, b]$, the right-sided RL fractional integral is

$$(I_{b-}^{\alpha} f)(t) = \frac{1}{\Gamma(\alpha)} \int_t^b (\tau - t)^{\alpha-1} f(\tau) d\tau. \quad (228)$$

This is useful in boundary-value problems and symmetric formulations.

59 Fractional integral as convolution

59.1 Convolution form

Define

$$g_{\alpha}(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}, \quad t > 0.$$

Then

$$(I^{\alpha} f)(t) = (g_{\alpha} * f)(t) := \int_0^t g_{\alpha}(t - \tau) f(\tau) d\tau. \quad (229)$$

59.2 Laplace transform rule

Let $F(s) = \mathcal{L}\{f(t)\}(s)$. Then

$$\mathcal{L}\{I^{\alpha} f\}(s) = s^{-\alpha} F(s). \quad (230)$$

This is one of the most useful operational formulas.

60 Examples

60.1 Fractional integral of a constant

Let $f(t) = 1$. Then

$$(I^{\alpha} 1)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} d\tau = \frac{t^{\alpha}}{\Gamma(\alpha + 1)}. \quad (231)$$

60.2 Fractional integral of a power

Let $f(t) = t^{\mu}$ with $\mu > -1$. Then

$$(I^{\alpha} t^{\mu})(t) = \frac{\Gamma(\mu + 1)}{\Gamma(\mu + \alpha + 1)} t^{\mu + \alpha}. \quad (232)$$

60.3 Fractional integral of an exponential (series form)

Using $e^{at} = \sum_{n=0}^{\infty} \frac{(at)^n}{n!}$ and linearity:

$$I^{\alpha} e^{at} = \sum_{n=0}^{\infty} \frac{a^n}{n!} I^{\alpha} t^n = \sum_{n=0}^{\infty} \frac{a^n}{n!} \frac{\Gamma(n + 1)}{\Gamma(n + \alpha + 1)} t^{n + \alpha}. \quad (233)$$

61 Important remarks

61.1 Nonlocality / memory

Because

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau,$$

the value at time t depends on the full past history $0 \leq \tau \leq t$ with a power-law weight. This is the origin of memory effects in fractional models. :contentReference[oaicite:2]index=2

61.2 Kernel behavior

For $0 < \alpha < 1$, the kernel behaves like $(t - \tau)^{\alpha-1}$ which is integrable but singular near $\tau = t$:

$$(t - \tau)^{\alpha-1} \rightarrow \infty \quad \text{as } \tau \rightarrow t^-.$$

So recent history has the strongest weight, but all past contributes.

62 High-yield summary

- RL fractional integral ($\alpha > 0$):

$$(I^\alpha f)(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t - \tau)^{\alpha-1} f(\tau) d\tau.$$

- Semigroup property:

$$I^\alpha I^\beta = I^{\alpha+\beta}.$$

- Laplace transform:

$$\mathcal{L}\{I^\alpha f\} = s^{-\alpha} F(s).$$

- Examples:

$$I^\alpha 1 = \frac{t^\alpha}{\Gamma(\alpha + 1)}, \quad I^\alpha t^\mu = \frac{\Gamma(\mu + 1)}{\Gamma(\mu + \alpha + 1)} t^{\mu+\alpha}.$$

63 Fractional calculus using discrete calculus

63.1 Discrete derivative and finite differences

Start from the forward difference operator:

$$(\Delta_h f)(t) := f(t+h) - f(t), \quad (234)$$

and the backward difference:

$$(\nabla_h f)(t) := f(t) - f(t-h). \quad (235)$$

For integer order $n \in \mathbb{N}$, the n th backward difference is

$$\nabla_h^n f(t) = \sum_{k=0}^n (-1)^k \binom{n}{k} f(t - kh). \quad (236)$$

The classical derivative is recovered as

$$f'(t) = \lim_{h \rightarrow 0^+} \frac{\nabla_h f(t)}{h}. \quad (237)$$

64 Grünwald–Letnikov (GL) fractional derivative

64.1 Definition

The Grünwald–Letnikov fractional derivative extends the finite-difference form from integer n to real α by replacing $\binom{n}{k}$ with $\binom{\alpha}{k}$:

$$(D_{GL}^\alpha f)(t) = \lim_{h \rightarrow 0^+} \frac{1}{h^\alpha} \sum_{k=0}^{\lfloor t/h \rfloor} (-1)^k \binom{\alpha}{k} f(t - kh). \quad (238)$$

Here

$$\binom{\alpha}{k} = \frac{\Gamma(\alpha+1)}{\Gamma(k+1)\Gamma(\alpha-k+1)}. \quad (239)$$

64.2 Consistency with integer derivatives

If $\alpha = n \in \mathbb{N}$, then

$$\binom{n}{k} = 0 \quad \text{for } k > n,$$

so the sum truncates and GL reduces to the standard n th derivative limit.

64.3 Alternative coefficient form

Define coefficients

$$c_k^{(\alpha)} := (-1)^k \binom{\alpha}{k}. \quad (240)$$

Then

$$(D_{GL}^\alpha f)(t) = \lim_{h \rightarrow 0^+} \frac{1}{h^\alpha} \sum_{k=0}^{\lfloor t/h \rfloor} c_k^{(\alpha)} f(t - kh). \quad (241)$$

64.4 Recursive computation of coefficients (useful numerically)

The binomial coefficients satisfy a recursion:

$$c_0^{(\alpha)} = 1, \quad c_k^{(\alpha)} = \left(1 - \frac{\alpha + 1}{k}\right) c_{k-1}^{(\alpha)} \quad (k \geq 1). \quad (242)$$

This allows efficient evaluation of the GL derivative on a grid.

65 Relation to non-locality (memory)

65.1 GL derivative is explicitly nonlocal

The GL derivative at time t uses *all past values*

$$f(t), f(t-h), f(t-2h), \dots, f(t - \lfloor t/h \rfloor h).$$

So the derivative at t depends on the entire history on $[0, t]$.

This is the discrete analogue of Volterra-type memory integrals emphasized in nonlocality theory.
:contentReference[oaicite:1]index=1

65.2 Memory kernel behavior (long-range influence)

For non-integer α , the coefficients $c_k^{(\alpha)}$ do not truncate. They decay slowly with k (power-law type), so distant past contributes.

Hence fractional differentiation encodes **long memory**:

$$(D^\alpha f)(t) \text{ depends on } \{f(\tau) : 0 \leq \tau \leq t\}. \quad (243)$$

65.3 Local limit

In the limit $\alpha \rightarrow 1$, the GL derivative approaches the ordinary derivative (which is local in the continuum limit):

$$\lim_{\alpha \rightarrow 1} D_{GL}^\alpha f(t) = f'(t). \quad (244)$$

66 GL derivative vs Riemann–Liouville / Caputo

66.1 Equivalence (conceptual)

Under suitable smoothness assumptions, the GL derivative coincides with the Riemann–Liouville derivative:

$$D_{GL}^\alpha f(t) = D_{RL}^\alpha f(t), \quad (245)$$

so GL provides a **discrete definition** of the RL derivative.

66.2 Why GL is important in applications

- It gives an immediate **finite-difference discretization** for fractional PDEs/ODEs.
- It makes nonlocality obvious: the derivative is a weighted sum of the entire past.
- It naturally leads to numerical “history” terms (memory cost grows with time).

67 Practical discretization formula

On a grid $t_n = nh$, define $f_n = f(t_n)$. A common GL approximation is:

$$(D^\alpha f)(t_n) \approx \frac{1}{h^\alpha} \sum_{k=0}^n c_k^{(\alpha)} f_{n-k}, \quad c_k^{(\alpha)} = (-1)^k \binom{\alpha}{k}. \quad (246)$$

Computational cost. This is $O(n)$ work per time step if done directly, since all history values appear. This is the numerical manifestation of **nonlocal memory**.

68 High-yield summary

- Finite differences define integer derivatives; GL extends them to fractional order.
- Grünwald–Letnikov derivative:

$$D_{GL}^\alpha f(t) = \lim_{h \rightarrow 0^+} \frac{1}{h^\alpha} \sum_{k=0}^{\lfloor t/h \rfloor} (-1)^k \binom{\alpha}{k} f(t - kh).$$

- Coefficients use Gamma functions:

$$\binom{\alpha}{k} = \frac{\Gamma(\alpha + 1)}{\Gamma(k + 1)\Gamma(\alpha - k + 1)}.$$

- GL derivative is **explicitly nonlocal** (depends on entire history).
- Numerically:

$$D^\alpha f(t_n) \approx h^{-\alpha} \sum_{k=0}^n c_k^{(\alpha)} f_{n-k},$$

so fractional dynamics naturally carries memory cost.

69 Fractional friction model

69.1 Motivation: friction with memory

Ordinary (local) linear friction is proportional to instantaneous velocity:

$$F_{\text{fr}}(t) = -\gamma \dot{x}(t). \quad (247)$$

This is **local in time**.

In many media (viscoelastic, complex fluids, disordered environments), the frictional response depends on the **history** of motion. This is modeled by a **nonlocal** friction law:

$$F_{\text{fr}}(t) = - \int_0^t M(t - \tau) \dot{x}(\tau) d\tau, \quad (248)$$

where $M(t)$ is a memory kernel (Volterra-type causal operator).

69.2 Fractional friction kernel

A common long-memory choice is a power-law kernel:

$$M(t) \propto t^{-\alpha}, \quad 0 < \alpha < 1. \quad (249)$$

This leads naturally to a **fractional derivative** of the velocity.

70 Fractional friction equation of motion

70.1 Fractional damping force

A standard fractional friction model writes the friction force as

$$F_{\text{fr}}(t) = -\gamma_\alpha {}^C D_t^\alpha x(t), \quad 0 < \alpha < 1, \quad (250)$$

or sometimes in terms of a fractional derivative of velocity:

$$F_{\text{fr}}(t) = -\gamma_\alpha {}^C D_t^\alpha \dot{x}(t). \quad (251)$$

(Exact form depends on the physical convention and units.)

70.2 Free fractional relaxation (prototype model)

The simplest fractional friction dynamics is the fractional relaxation equation:

$${}^C D_t^\alpha x(t) + \lambda x(t) = 0, \quad 0 < \alpha \leq 1. \quad (252)$$

For $\alpha = 1$ this reduces to ordinary exponential relaxation:

$$\dot{x} + \lambda x = 0.$$

70.3 Solution: Mittag–Leffler relaxation

The solution is expressed using the Mittag–Leffler function:

$$x(t) = x(0) E_\alpha(-\lambda t^\alpha), \quad (253)$$

where

$$E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + 1)}. \quad (254)$$

Key property. For $0 < \alpha < 1$, relaxation is typically **slower than exponential** (power-law tail behavior at late times), reflecting memory.

71 Relaxation vs amplification

71.1 Relaxation (stable decay)

Fractional relaxation corresponds to a negative “effective rate” in the argument:

$${}^C D_t^\alpha x(t) + \lambda x(t) = 0, \quad \lambda > 0. \quad (255)$$

Solution:

$$x(t) = x(0) E_\alpha(-\lambda t^\alpha).$$

For $\lambda > 0$ this decays to 0 (stable).

71.2 Amplification (growth / instability)

If the sign is reversed,

$${}^C D_t^\alpha x(t) - \lambda x(t) = 0, \quad \lambda > 0, \quad (256)$$

then

$$x(t) = x(0) E_\alpha(+\lambda t^\alpha), \quad (257)$$

which grows in time (unstable amplification).

71.3 Physical interpretation

- **Relaxation:** friction/memory dissipates perturbations.
- **Amplification:** energy is effectively injected or feedback dominates.

72 Main properties of fractional friction models

72.1 Nonlocality (memory)

Fractional derivatives depend on the entire past:

$${}^C D_t^\alpha x(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-\tau)^{-\alpha} \dot{x}(\tau) d\tau. \quad (258)$$

Thus fractional friction is inherently **history-dependent**.

72.2 Exponential vs Mittag–Leffler decay

- Classical damping ($\alpha = 1$): $x(t) \sim e^{-\lambda t}$.
- Fractional damping ($0 < \alpha < 1$): $x(t) \sim E_\alpha(-\lambda t^\alpha)$ (decay is typically slower, often with algebraic tail).

72.3 Frequency response (qualitative)

In Fourier/Laplace space, fractional differentiation behaves like a power:

$${}^C D_t^\alpha \sim s^\alpha. \quad (259)$$

Hence fractional friction produces frequency-dependent dissipation:

$$\text{effective damping} \propto \omega^\alpha. \quad (260)$$

This matches many complex media where damping is not constant in frequency.

72.4 Interpolation between elastic and viscous behavior

Fractional friction often interpolates between:

- purely elastic response (instantaneous / no dissipation),
- purely viscous response (local friction proportional to $\dot{x}$),

by introducing intermediate scaling controlled by α .

72.5 Initial conditions

Using Caputo derivatives, initial conditions remain classical:

$$x(0) = x_0, \quad \dot{x}(0) = v_0 \quad (\text{if needed}). \quad (261)$$

This is one reason Caputo form is preferred in physical models.

73 High-yield summary

- Fractional friction models damping with memory (nonlocal in time).
- Prototype fractional relaxation equation:

$${}^C D_t^\alpha x + \lambda x = 0, \quad 0 < \alpha \leq 1.$$

- Solution uses Mittag–Leffler function:

$$x(t) = x(0) E_\alpha(-\lambda t^\alpha).$$

- Relaxation: $-\lambda$ argument $\Rightarrow$ decay (stable).
- Amplification: $+\lambda$ argument $\Rightarrow$ growth (unstable).
- Main features: long memory, slower-than-exponential decay, frequency-dependent damping.

74 Fractional oscillator model

74.1 Classical harmonic oscillator

The classical oscillator is

$$\ddot{x}(t) + \omega^2 x(t) = 0, \quad (262)$$

with solutions $x(t) = A \cos(\omega t) + B \sin(\omega t)$.

74.2 Fractional harmonic oscillator (general idea)

A fractional oscillator replaces integer derivatives by fractional ones. A standard prototype is

$$\boxed{{}^C D_t^{2\alpha} x(t) + \omega^2 x(t) = 0, \quad 0 < \alpha \leq 1,} \quad (263)$$

where ${}^C D_t^{2\alpha}$ is the Caputo fractional derivative. For $\alpha = 1$ this reduces to the classical oscillator.

74.3 Caputo fractional trigonometric functions (Herrmann)

Herrmann constructs two linearly independent solutions of the fractional oscillator and denotes them in analogy to sine/cosine:

$$\boxed{C \sin(\alpha, t) = \sum_{n=0}^{\infty} (-1)^n \frac{t^{(2n+1)\alpha}}{\Gamma(1 + (2n+1)\alpha)},} \quad (264)$$

$$\boxed{C \cos(\alpha, t) = \sum_{n=0}^{\infty} (-1)^n \frac{t^{2n\alpha}}{\Gamma(1 + 2n\alpha)}.} \quad (265)$$

These satisfy the key derivative relations (Caputo derivative):

$$\boxed{{}^C D_t^\alpha C \sin(\alpha, \omega t) = \omega^\alpha C \cos(\alpha, \omega t),} \quad (266)$$

$$\boxed{{}^C D_t^\alpha C \cos(\alpha, \omega t) = -\omega^\alpha C \sin(\alpha, \omega t).} \quad (267)$$

Herrmann emphasizes these as the natural fractional analogues of trig functions. :contentReference[oaicite:4]index=4

74.4 Mittag–Leffler representation

The above series are special cases of Mittag–Leffler functions:

$$C \cos(\alpha, t) = E(2\alpha, -t^{2\alpha}), \quad C \sin(\alpha, t) = t^\alpha E(2\alpha, 1 + \alpha, -t^{2\alpha}). \quad (268)$$

Mittag–Leffler functions play the role of exponentials in fractional dynamics. :contentReference[oaicite:5]index=5

74.5 Intrinsic damping (relaxation of oscillations)

A crucial physical observation in Herrmann:

For $\alpha < 1$, fractional oscillator solutions behave like *damped oscillations* even without adding an external friction term.

So the damping is **intrinsic** to the fractional time evolution, not due to external forces. :contentReference[oaicite:6]index=6 :contentReference[oaicite:7]index=7

74.6 Definition-independence (robust qualitative behavior)

Herrmann compares solutions for different fractional derivative definitions and notes:

- for $\alpha \lesssim 1$, oscillations have similar frequency across definitions,
- the main differences appear in the number of zeros (important for boundary problems).

:contentReference[oaicite:8]index=8

75 Fractional wave equations

75.1 Classical wave equation

In 3D:

$$\left(\Delta - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right) \psi(\mathbf{x}, t) = 0. \quad (269)$$

75.2 Fractional extension (spatial fractionalization)

Herrmann proposes a fractional wave equation by replacing spatial second derivatives with fractional ones (time kept second order):

$$\left(\frac{d^{2\alpha}}{dx^{2\alpha}} + \frac{d^{2\alpha}}{dy^{2\alpha}} + \frac{d^{2\alpha}}{dz^{2\alpha}} - \mu^2 \frac{d^2}{dt^2} \right) \psi(x, y, z, t) = 0. \quad (270)$$

Here μ carries dimensions $[s/m^\alpha]$. :contentReference[oaicite:9]index=9

75.3 Separation of variables

Using the product ansatz

$$\psi(x, y, z, t) = X(x)Y(y)Z(z)T(t), \quad (271)$$

the PDE reduces to uncoupled ODEs:

$$\frac{d^{2\alpha}}{dx^{2\alpha}} X = -\kappa_x^2 X, \quad \frac{d^{2\alpha}}{dy^{2\alpha}} Y = -\kappa_y^2 Y, \quad \frac{d^{2\alpha}}{dz^{2\alpha}} Z = -\kappa_z^2 Z, \quad (272)$$

$$\frac{d^2}{dt^2} T = \frac{\omega^2}{\mu^2} T. \quad (273)$$

The separation constants satisfy:

$$\omega = \mu \sqrt{\kappa_x^2 + \kappa_y^2 + \kappa_z^2}. \quad (274)$$

:contentReference[oaicite:10]index=10 :contentReference[oaicite:11]index=11

75.4 Boundary conditions viewpoint

Wave problems are often determined by boundary conditions (Dirichlet/Neumann/mixed), not initial conditions. Herrmann highlights that this makes Riemann-type solutions (even if singular at $t = 0$) still useful in wave contexts. :contentReference[oaicite:12]index=12

76 Parity, regularization, and localization

76.1 Why parity becomes nontrivial

Fractional powers like x^α are multivalued for $x < 0$:

$$(-x)^\alpha = (-1)^\alpha |x|^\alpha$$

which causes issues for defining physical wave solutions on domains containing $x < 0$. :contentReference[oaicite:13]index=13

76.2 Herrmann's parity-regularized coordinate

To preserve parity behavior, Herrmann introduces a regularized coordinate

$$\hat{x}^\alpha := \text{sign}(x)|x|^\alpha, \quad (275)$$

and replaces

$$\frac{d}{dx} \longrightarrow \frac{d^\alpha}{d\hat{x}^\alpha} =: \hat{D}_x^\alpha. \quad (276)$$

This yields a *regularized fractional wave equation* with proper parity properties:

$$\left((\hat{D}_x^\alpha)^2 + (\hat{D}_y^\alpha)^2 + (\hat{D}_z^\alpha)^2 - \mu^2 \frac{d^2}{dt^2} \right) \psi(x, y, z, t) = 0. \quad (277)$$

Free solutions then transform under parity similarly to the classical ($\alpha = 1$) case. :contentReference[oaicite:14]index=14

76.3 Localization (qualitative meaning)

In fractional wave/quantum problems, solutions may exhibit stronger localization due to:

- nonlocal spatial operators (long-range coupling),
- altered nodal structure (different zero distributions),
- modified dispersion relations (effective $k^{2\alpha}$ scaling).

In Herrmann's framework, the number/location of zeros is a major qualitative difference between fractional and classical oscillatory solutions. :contentReference[oaicite:15]index=15

77 Relaxation vs localization

77.1 Relaxation (temporal damping)

Fractional oscillators with $\alpha < 1$ show **relaxation**:

- oscillations decay even without explicit friction,
- decay is governed by Mittag-Leffler-type behavior,
- memory effects are intrinsic to fractional derivatives.

This is one of the central physical signatures of fractional time dynamics. :contentReference[oaicite:16]index=16

77.2 Localization (spatial fractionalization)

Spatial fractional wave equations modify eigenmodes. Under boundary conditions, the spatial fractional eigenfunctions can:

- deviate from ordinary sines/cosines,
- show different node counts and spacing,
- depend more sensitively on parity/regularization choices.

These effects can be interpreted as a form of localization/shape deformation compared to classical wave modes. :contentReference[oaicite:17]index=17 :contentReference[oaicite:18]index=18

78 Main properties (high-yield)

- Fractional oscillator solutions are expressed via Mittag-Leffler functions (the fractional analogue of exponentials). :contentReference[oaicite:19]index=19
- For $\alpha < 1$, the fractional harmonic oscillator behaves like a **damped oscillator** with **intrinsic absorption** (no external friction needed). :contentReference[oaicite:20]index=20
- Fractional wave equations can be formed by fractionalizing spatial derivatives, keeping time second order. :contentReference[oaicite:21]index=21
- Separation of variables still works and yields fractional eigenvalue ODEs. :contentReference[oaicite:22]index=22
- Parity and negative arguments require care; Herrmann introduces a regularization $\hat{x}^\alpha = \text{sign}(x)|x|^\alpha$ to preserve parity symmetry. :contentReference[oaicite:23]index=23
- Boundary conditions are central for wave problems; fractional solutions may differ strongly in zero structure (important for mode spectra). :contentReference[oaicite:24]index=24